

Happiest Health · Digital Twin Challenge 2026

ThermoTwin

A hypertension digital twin that knows how your medicines handle
India's heat

[Team name] · [College / incubator]

Heat is hiding inside heart deaths

India's heat toll is massively under-counted, and hypertension sits at the centre of it

14

official heatstroke
deaths in India, 2025

~30,000

estimated excess deaths
from a single five-day
heatwave

315 M

Indian adults living with
hypertension

~380 M

workers whose jobs
expose them to heat

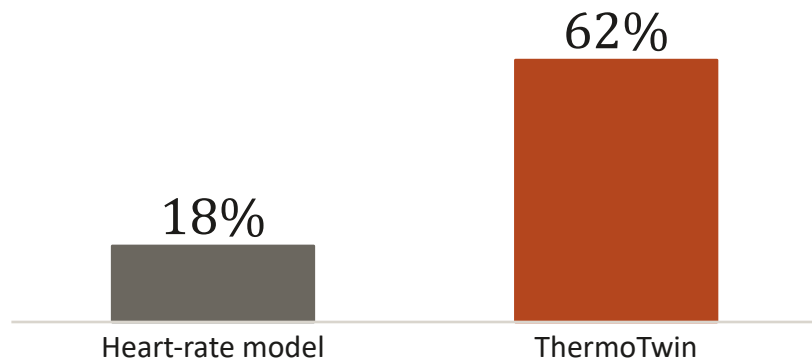
Many heat deaths are recorded as cardiac arrest. Blood-pressure medicines change how the body handles heat, but no tool connects a patient's record to today's weather.

The blind spot: beta-blockers fool wearables

Heat models read core temperature from heart rate, and beta-blockers stop heart rate rising

- Wearable heat models assume heart rate rises with core temperature.
- Beta-blockers blunt that rise, so estimates read low for exactly the patients at risk.
- ThermoTwin reads the medication record and corrects the physiology before estimating.

Danger minutes ($\geq 38^\circ\text{C}$) caught, β -blocker patients



Tested on real heat-trial data (PROSPIE) with a simulated beta-blocker effect on heart rate.

ThermoTwin in one slide

A digital twin of a hypertensive outdoor worker, updated minute by minute

- | | | |
|---|---------------|--|
| 1 | Condition | Hypertension, with the medicines that treat it (β -blockers, ACE-i/ARB, CCB, diuretics) |
| 2 | Adverse event | Core temperature ≥ 38 °C: now, and forecast 60 minutes ahead |
| 3 | Inputs | Static EHR + 1-minute wearable heart rate and activity + weather |
| 4 | Users | PHC doctor (dashboard, pre-summer review) and worker (rest alert) |

ThermoTwin architecture

Hybrid physics + ML twin: the medication record changes the physiology model, not just the score

DATA STREAMS

Static EHR (synthetic)

BP meds by class, eGFR history, NSAIDs, age, BMI

Wearable stream

1-min heart rate + activity (JOS-3 simulated shifts)

Weather

Open-Meteo archive: Delhi heatwave, May 2024

Real heat trials

PROSPIE (22 people, 99 trials): validation only

TWIN ENGINE (PYTHON)

1 ECTemp Kalman filter

Core temperature from heart rate

2 Medication correction

β -blocker: un-blunt HR, widen noise

3 Personal bias learning

Thermometer readings from past shifts

4 60-min forecaster

Gagge physics + gradient boosting

SERVICE

FastAPI

/patients
/timeline
/whatif
/review

Review rules

Triple whammy, eGFR decline, β -blocker masking, heat strain

USERS

Dashboard

React: live twin, what-if, review

PHC doctor

Pre-summer medication review

Deploy: Docker Compose (non-root API with health check · nginx serves dashboard and proxies /api) · Quality: pytest suite, participant-held-out real-data validation, multi-seed evaluation

How the twin thinks

Physics first, then personalisation, then forecasting

1

Estimate

ECTemp Kalman filter turns 1-minute heart rate into core temperature (Buller et al. 2013).

2

Correct

If the EHR lists a β -blocker, un-blunt heart rate with population priors and widen uncertainty.

3

Personalise

Learn each worker's bias from a few thermometer readings on earlier shifts.

4

Forecast

Gagge two-node physics + gradient boosting predict $\geq 38^\circ\text{C}$ within 60 minutes.

Honest by design: ground truth comes from a different physiology model (JOS-3) than the twin uses, and each patient's true drug response is hidden from it.

Validated on real heat trials

PROSPIE: 22 participants, 99 trials, rectal core temperature, held-out participants

0.32 °C

core-temperature error of our
heart-rate engine on real people,
in line with the published ~0.3 °C

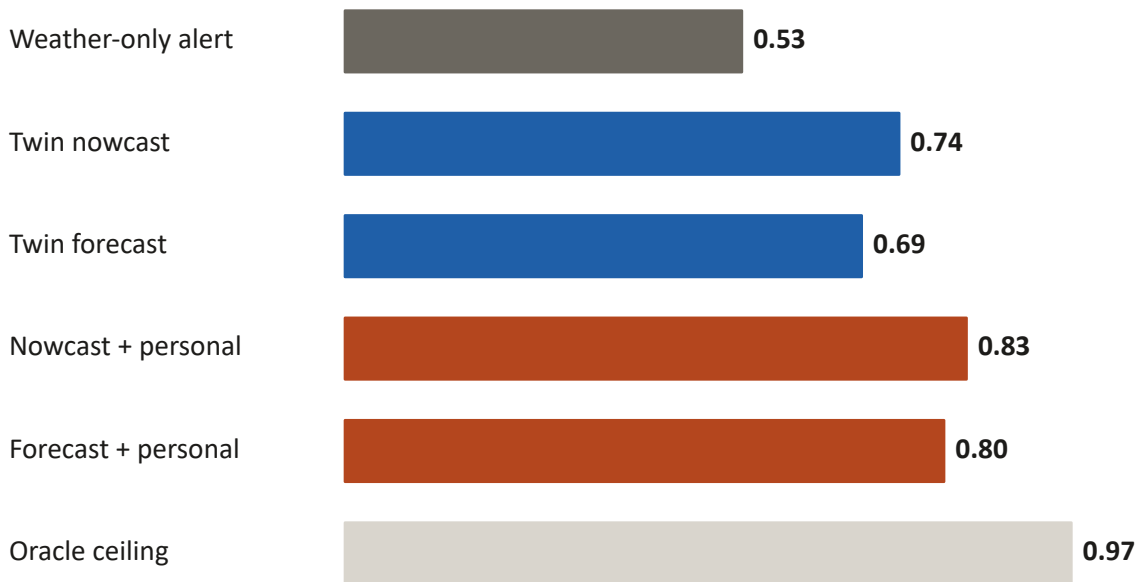
β-blocker scenario (real data)	Error	Danger caught	False alarms
Heart-rate model	0.63 °C	18%	0%
ThermoTwin (medication-aware)	0.39 °C	62%	8%
+ learned personal bias *	0.37 °C	79%	38%

** Separate run with a different random drug response: 0.44 → 0.37 °C and 59% → 79% caught, but false alarms rise 13% → 38%, a trade-off we report. Skin-temperature fusion gave no real gain, so we don't claim it.*

60-minute forecast on a real Delhi heatwave

200 simulated workers on May–June 2024 weather (peaks of 46 °C), tested on unseen patients, 3 seeds

AUROC, mean of 3 seeds (higher is better)



+0.11

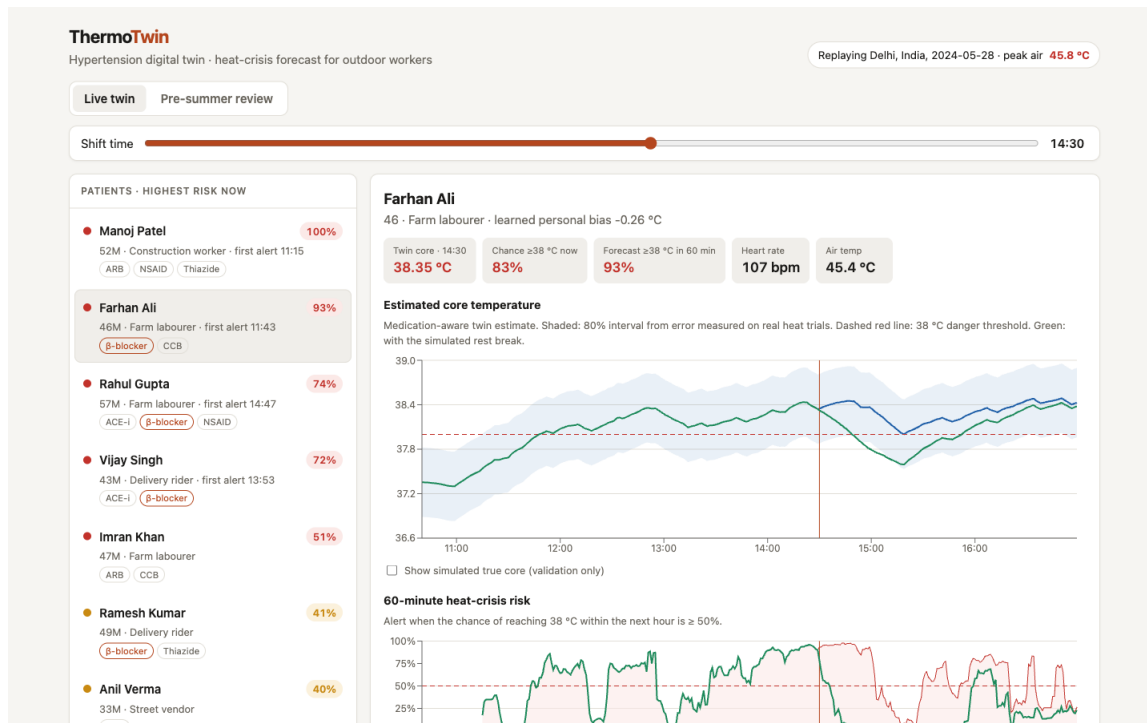
AUROC gain from learning each worker, consistent on every seed

31%

of danger episodes warned $\geq$ 30 min ahead (weather-only alert: 0%)

The clinician's live twin

Real 28 May 2024 Delhi replay: patients ranked by risk, estimate with an 80% interval sized from real-data error



Two numbers, never contradictory:
chance ≥ 38 °C now, and
forecast within 60 min

Medication chips
β-blockers highlighted: their
wearable readings are corrected

Validation toggle
overlay simulated true core
temperature

From prediction to action

What-if rest breaks for today; a medication review before summer

What if Farhan rests 30 min in the shade at 14:30?

What if they rest in the shade?

Re-runs this worker's twin with an extra rest break starting at the selected shift time.

Rest for from 14:30 Simulate rest break Clear

Peak twin core

38.49 °C → 38.44 °C

Alert minutes

207 → 141

Minutes truly ≥38 °C

39 → 16

True minutes ≥ 38 °C fall from 39 to 16; alert minutes from 207 to 141.

Pre-summer review ranks patients with cited rules: triple-whammy kidney risk (BMJ 2013), falling eGFR (KDIGO 2024), β -blocker masking, heat-illness drugs (2026 cohort) and predicted heat strain.

ThermoTwin
Hypertension digital twin · heat-crisis forecast for outdoor workers

Replaying Delhi, India, 2024-05-28 · peak air 45.8 °C

Pre-summer medication & heat review
Suggestions for clinician review before the heat season. They combine medication and kidney records with each twin's replay of the 28 May 2024 Delhi heatwave. Priority is a triage heuristic, not a validated clinical score.

High priority
Manoj Patel
S2 · Construction worker
ARB · NSAID · Thiazide
eGFR 84 → 80 (-5%) Heat strain (replay) 284 min ≥ 38 °C

Triple-whammy kidney risk
Diuretic + ACE inhibitor/ARB + NSAID; dehydration in heat adds to the risk.
→ Review NSAID use (consider paracetamol); check creatinine and potassium before the heat season.
Lapi et al., BMJ 2013;346:e8529

Twin predicts heat strain on heatwave days
About 284 min likely ≥ 38 °C core in the 28 May 2024 Delhi replay.
→ Advise avoiding peak hours (12:00 to 15:00) and a shade break every hour.
NDMA heatwave guidelines; ThermoTwin replay

Higher heat-illness risk in heatwaves
On arb.
→ Counsel on hydration, shade and what to do if dehydrated (sick-day guidance).
Antihypertensives and heat-related illness during heatwaves, 2026 (doi:10.1002/pds.70447)

High priority
Vijay Singh
A3 · Delivery rider
ACE-i **β -blocker**
eGFR 103 → 84 (-18%) Heat strain (replay) 270 min ≥ 38 °C

Twin predicts heat strain on heatwave days
About 270 min likely ≥ 38 °C core in the 28 May 2024 Delhi replay.
→ Advise avoiding peak hours (12:00 to 15:00) and a shade break every hour.
NDMA heatwave guidelines; ThermoTwin replay

Kidney function falling
eGFR 103 → 84 mL/min/1.73 m² in a year (-18%).
→ Recheck eGFR and electrolytes before summer; review diuretic dose and hydration advice.
KDIGO 2024 CKD guideline (rapid progression > 5 mL/min/1.73 m²/yr)

Heart rate under-reports heat strain
Beta-blockers blunt the heart-rate rise that wearables use to estimate heat.
→ Do not rely on heart-rate alerts alone; schedule fixed shade breaks.

Printable list for the PHC doctor · suggestions for clinician review, not orders

What makes ThermoTwin different

Compared with a typical risk-score prototype

	Typical prototype	ThermoTwin
Role of the EHR	Extra model features	Changes the physiology model
What “twin” means	Classifier + dashboard	Kalman filter + physics + personal learning
Validation	Synthetic only	Real heat trials, held-out participants
Honesty	Single accuracy number	Baselines, oracle ceiling, 3 seeds, negative results
Action	“Risk is high”	What-if rest breaks + medication review
India fit	Generic	Real Delhi heatwave, PHC workflow, outdoor workers

What we have not proven yet

Stated plainly, with the next step for each

Simulated patients

Only the weather and validation data are real.

→ Pilot with a PHC hypertension clinic under ethics approval

Drug effect simulated

PROSPIE volunteers were healthy; the β -blocker response was applied to their heart rate.

→ Validate on real β -blocker users

Gap to the oracle

Forecast AUROC 0.80 vs 0.97 if current core were known.

→ Better per-person heart-rate models

False alarms

Personal learning raises detection but also alerts.

→ Calibrate thresholds with clinicians

ThermoTwin

The patient's record should change how we read their body in the heat.

Run it: `docker compose up --build` → <http://localhost:8080>
Code, results and evaluation reports: [GitHub repository](#) (MIT licence)

Research prototype on simulated patients and real weather. Not a medical device.